# Replication of the orthogonal latent-mean contrasts in Jeffries (2024)

28 October 2026

**Abstract**

Do the latent-mean contrasts in Jeffries (2024) sit on the printed scale −0.40, −0.20, 0, and do the rejection rates in Tables 4.1 and 4.3 follow from that model? Jeffries defines three codes for four groups and reports multigroup and MIMIC averages near those three numbers when the generating means are 1, 0.6, 0.8, and 0.4. Fitting the marker model to the population that lavaan [2] builds from the generation string returns exactly −0.40, −0.20, and 0. The unbalanced power rates at $N = 240$ come out 49.35, 21.28, and 8.51 against the printed 49.23, 21.08, and 7.77. Of thirty rejection-rate comparisons at 10,000 replicates, three miss an uncorrected two-standard-error band and none miss Bonferroni or Holm. A later draw with simsem 0.5.16 [3] moves two of those three inside the band and leaves one: 6.46 against 7.41.

## 1 The question, and the headline

This section answers whether the printed contrasts and the printed rejection rates come from the model in the dissertation. The contrasts do. Most of the rates do. One rate, one average, and the MIMIC convergence counts are still open.

Jeffries (2024), https://digitalcommons.unl.edu/dissunl/216/, studies one factor, four indicators, and four groups. The true-difference means are 1, 0.6, 0.8, and 0.4. The null means are all 1. Sample sizes are 240, 480, and 960, with group shares 25/25/25/25, 35/35/15/15, and 45/45/5/5. The fitting program named in the text is lavaan 0.6.17. The public analysis scripts are at https://osf.io/q9yu8/.

The population fit of the marker model on the parsed generation string is −0.40, −0.20, 0. The same marker model on a population in which every loading is 0.70 is −0.28, −0.14, 0. The printed tables match the first of those. At $N = 240$ and 10,000 replicates, the unbalanced power rates match the page, and a Bonferroni or Holm correction of the thirty rate comparisons leaves no miss. The rate that remains outside an uncorrected band is the MIMIC model, shares 45/45/5/5, null means, contrast 2.

The original work is the design, the codes, and the printed tables. The calculations below are ours: the moment fits, the Monte Carlo, and the code identities. The dissertation's percent columns are one Wald test per dataset at $\alpha = 0.05$. That is the design of those columns. Multiplicity is applied only to our check that a printed rate and our rate agree.

## 2 What we replicated

This section answers which printed claims agree with a direct calculation. Table 1 is that comparison. "Match" means the same number, the same population target, or a Monte Carlo gap no larger

than two combined standard errors. "Partial" means one piece holds and another does not. "No match" means a different scale. "Close" means a gap sitting on the edge of that two-standard-error band. The combined standard error uses the printed count of 10,000 and our count of converged replicates.

| Claim | Published | Ours | Verdict |
|---|---|---|---|
| Defined contrasts on $(1, 3/5, 4/5, 2/5)$ | $-0.40,\ -0.20,\ 0$ | $-2/5,\ -1/5,\ 0$ | match |
| Unweighted products of the Table 2.2 codes | Equation (2.4) is zero | Proved zero | match |
| Imbalance makes the set nonorthogonal | The set becomes nonorthogonal | Only code 1 with code 3 fails | partial |
| Marker model on the parsed generation population | Averages near $-0.40,\ -0.20,\ 0$ | Population fit $-0.40,\ -0.20,\ 0$ | match |
| Every loading 0.70, then a marker loading of 1 | The same averages | Population fit $-0.28,\ -0.14,\ 0$ | no match |
| Power, Table 4.1, $N = 240$, shares 45/45/5/5 | 49.23, 21.08, 7.77 | 49.35, 21.28, 8.51 | match |
| Null rates, Table 4.3, same cell | 8.13, 8.59, 7.73 | 8.10, 8.33, 8.22 | match |
| Thirty rejection rates, uncorrected | Each percent is one test at $\alpha = 0.05$ | 3 of 30 outside two combined SEs | partial |
| Same thirty rates, Bonferroni and Holm | Those tests are a single-test design | 0 of 30 outside the band | match |
| Multigroup convergence, Table 4.5, that cell | 89.29% | 89.82% of 10,000 | match |

Table 1: Published claims against this replication. Monte Carlo rows use 10,000 replicates at $N = 240$. The population rows use moment matrices.

The contrast row is equations (3.3)–(3.5) on the printed means, proved in Lean [4] at 123. The product row is equation (2.4), proved at 103. The imbalance row is partial because sizes $(a, a, b, b)$ leave two of the three weighted products at zero and set the other to $2(b - a)$, proved at 68. At $N = 240$ that product is $-96$ in condition 2 and $-192$ in condition 3. The third contrast is zero while the first two means differ, proved at 127. The treatment-coefficient form of the same contrasts is proved at 159. The marker identity at 131, which multiplies the contrasts by 7/10, is the every-loading-0.70 reading in the table. It is the population that misses $-0.40$.

The power row is the multigroup model in the smallest groups. The largest $|z|$ among those three rates is 1.85. The averages in that cell are $-0.403$, $-0.215$, and 0.0050 against $-0.404$, $-0.218$, and 0.0054. The thirty-rate rows use one normal test per rate. The uncorrected cutoff is $|z| > 1.96$, and the two-standard-error band used in the first screen is the same rule with 2 in place of 1.96. Bonferroni on thirty tests uses $z = \Phi^{-1}(1 - 0.05/30) = 2.94$, and the two-sided threshold is $z = \Phi^{-1}(1 - 0.05/60) = 3.14$. Holm sorts the thirty two-sided $p$-values and stops at the first failure. At 10,000 replicates the smallest of those $p$-values is 0.0083, above $0.05/30 = 0.00167$, so Holm stops immediately.

A binomial test of three misses in thirty trials at $p = 0.05$ has upper tail 0.19. That figure treats the thirty indicators as independent. The three contrasts in a cell share replicates, so the count is dependent inside each cell and the binomial probability is the independent-trials calculation. The

expected count under independence is 1.5. Three misses is an ordinary count for that band.

# 3 What gave us trouble, and how it was resolved

This section answers why an earlier replay sat on −0.28 and why a short Monte Carlo looked as if many rates had missed.

## 3.1 The scale −0.28 against −0.40

The generation line is `eta =~ NA*i1 + 0.7*i1 + 0.7*i2 + 0.7*i3 + 0.7*i4`. In lavaan 0.6.17 that line keeps a free loading on the first indicator and drops the duplicate 0.7. `simulateData` fills the free loading with 1. The other three loadings stay at 0.70. The factor variance in the syntax is 0.50 and the residual variances are 0.51. A large sample from that string has mean about 2.00 on the first indicator when the factor mean is 1, and means about 1.70 on the other three, which is $1 + 0.70 \times 1$.

The marker model fixes the first loading at 1. On this population the factor stays on the generating scale, and the three contrasts are −0.40, −0.20, and 0. If every loading is generated at 0.70 and the first is then fixed at 1, the same contrasts are rescaled by 0.70, which is −0.28, −0.14, and 0. The printed tables match the parsed string. The moment fits are `simulation/candidate_readings.R`, and the output is `simulation/out/candidate_readings.csv`.

> Latent variables were identified using the marker method approach, where the first indicator's loading was fixed to one ($\lambda_1 = 1$). Per recommendations from Hancock (1997), over both SEM forms, intercepts were modeled such that $\tau_{1_1} = \tau_{2_1} = \ldots \tau_{x_i}$. Further, the variance of all latent variables were freely estimated.

Figure 1: Marker identification. The first loading is fixed at 1, and the intercepts are equated. PDF page 90.

> model results for models of data that were explicitly generated to show latent mean differences (see Tables 4.1 and 4.2). As a reminder, the data-generated latent mean difference for Contrasts 1 through 3 were, on average, -0.40, -0.20, and 0 (i.e., a null difference), respectively. In the null model, the data-generated mean differences for Contrasts 1 through 3 were zero, on average, across all three contrasts. An average

Figure 2: The data-generating contrasts are −0.40, −0.20, and 0. PDF page 116.

The page writes `i1 ~ 1*`. That token does not parse in lavaan 0.6.17 or in simsem 0.5.17. The OSF generation script, and the null model on the same page, write `i1 ~ 1*1`. The Monte Carlo uses `1*1`.

The MIMIC line in `osf/analysis/mimic_model.R` has no free latent intercept, and lavaan then fixes that intercept at 0. The slopes on a large sample are about −0.15, 0.06, and 0.27. Adding

```
# ---- Models of N = 240 ----
pacman::p_load(here, tidyverse, simsem, lavaan, purrr)
set.seed(1313)
# ---- Significance Model ----
## ---- Model Syntax ----
# 4 item model
pop_model_cond1 <- "
eta =~ NA*i1 + 0.7*i1 + 0.7*i2 + 0.7*i3 + 0.7*i4
eta ~~ .5*eta
i1 ~~ 0.51*i1
i1 ~ 1*
i2 ~~ 0.51*i2
i2 ~ 1*1
i3 ~~ 0.51*i3
```

Figure 3: Appendix A. The first intercept is written `1*`. The loading line contains both `NA*i1` and `0.7*i1`.

the intercept returns −0.399, −0.200, and 0.000 on a sample of 100,000 per group (simulation/mimic_large.R, seed 7). The saved MIMIC summary is near −0.40, so the stored column is the free-intercept fit.

### 3.2 Six misses at 400 replicates, then three at 10,000

A 400-replicate screen of the ten $N = 240$ cells, multigroup and free-intercept MIMIC, true and null, put all 30 averages inside two combined standard errors and six rejection rates outside that band. The script is simulation/validate_gaps.R. The scores are simulation/out/gap_scores.csv. One of the six also missed Bonferroni and Holm: MIMIC, equal groups, null means, contrast 1, 2.26 against 5.18, $|z| = 3.77$. A binomial test of six misses in thirty trials at $p = 0.05$ has upper tail 0.0033 if the indicators are independent. The dependence caveat above applies to that figure as well.

The same ten cells at 10,000 replicates are simulation/validate_10k.R, with one fixed L'Ecuyer seed per cell in simulation/out/seeds_10k.csv. Five of the six short-run misses fall inside the uncorrected band, including the corrected miss, which comes out 5.01 against 5.18. Three rates remain outside:

| Cell | Contrast | Printed | At 10,000 | $\|z\|$ |
|---|---|---|---|---|
| MIMIC, 45/5, true difference | 2 | 23.37 | 22.11 | 2.12 |
| MIMIC, 45/5, null | 2 | 7.41 | 6.46 | 2.64 |
| MIMIC, equal groups, null | 3 | 5.32 | 4.53 | 2.59 |

Their averages match the page. Bonferroni and Holm leave none of the thirty rates outside. The scores are simulation/out/gap_scores_10k.csv.

### 3.3 The simsem 0.5.16 redraw, and the convergence rule

The dissertation names simsem 0.5.16. For a lavaan script, `sim()` in that version draws the rows with `lavaan::simulateData`, after setting the generator to L'Ecuyer-CMRG and the seed to the

`exportData` seed. `simulation/env_miss.R` reruns the miss cells that way, at seed 13131, with lavaan 0.6.17. The summary is `simulation/out/env_miss.csv`.

On that stream the three rates are 22.53 against 23.37 ($|z| = 1.42$), 4.86 against 5.32 ($|z| = 1.49$), and 6.46 against 7.41 ($|z| = 2.64$). Two fall inside the uncorrected band. The third is the same 6.46 the hand-written generator produced. The lavaan converged flag and the finite-standard-error rule selected the same replicates in every cell of this redraw. No captured warning contained the word "gradient." Requiring nonnegative variances dropped replicates only in the multigroup equal-null cell, 9,774 of 9,927, and that subset does not move the rates onto the page.

## 4 What is still outstanding

This section answers which comparisons are still open after the 10,000-replicate runs.

| Comparison | Printed | Ours |
|---|---|---|
| MIMIC, shares 45/45/5/5, null, contrast 2, percent significant | 7.41 | 6.46 ($\|z\| = 2.64$) |
| Multigroup, equal groups, null, contrast 3, average | −0.0021 | 0.0015 ($\|z\| = 1.98$) |
| MIMIC convergence, four $N = 240$ cells in the redraw | 10,000 of 10,000 | 9,988, 9,980, 9,964, 9,951 |
| Table 4.5, multigroup null, $N = 240$, condition 3 | 9,675 and 95.75% | 9,656 of 10,000 (96.56%) |
| MIMIC average, equal groups, true difference, contrast 1 | −0.4001 on the page | −0.399094 in the public file |

The 6.46 rate is outside the uncorrected band and inside Bonferroni and Holm on the family of thirty. Both generators produced it. The average 0.0015 uses the printed standard error 0.0013 as the standard deviation of one replicate. The nonnegative-variance subset of that cell is 0.0014, $|z| = 1.92$. An earlier stream of the same cell was 0.0023, $|z| = 2.40$. The gap sits on the boundary of the uncorrected band and inside $z = 2.94$.

The MIMIC rows of Table 4.5 print 10,000 of 10,000. The redraw is short of that count by 12 to 49 attempts. The converged flag and the finite-standard-error rule agree, so the shortfall is not a stricter filter on our side.

The multigroup null row at $N = 240$, condition 3, prints 9,675 and 95.75%. Dividing 9,675 by 10,000 gives 96.75. The rendered glyphs are 95.75, and the count is 9,675. Our count in the hand-written generator is 9,656 of 10,000, which is 96.56%. That rate is inside two combined standard errors of 96.75 and outside two combined standard errors of 95.75.

The page prints −0.4001 for the MIMIC average of contrast 1 at $N = 240$ in the equal true-difference cell. The public summary stores −0.399094. The significance count in that cell is 9,573 on the page and in the file.

## 5 What we tried on the outstanding gaps

This section answers whether the software, the public scripts, or another public copy of the code accounts for the gaps in Section 4. They do not. The open gaps are the size of a different random stream, or of a detail that is not in the public record. The next check is the author's data files and session record.

The dissertation names R 4.2.2, simsem 0.5.16, and lavaan 0.6.17, and `nlminb` with a maximum of 10,000 iterations. The operating system and `RNGkind` are absent from that description. The default since R 3.6.0, and the default here, is Mersenne-Twister, Inversion, and Rejection. `sim()` replaces that with L'Ecuyer-CMRG and then uses the `exportData` seed. simsem 0.5-16 was current from 2021-03-28 until 0.5-17 on 2025-04-02. lavaan 0.6-17 was published on 2023-12-20. The ten-cell Monte Carlo used R 4.3.3 [1]. R 4.2.2 was not installed. The L'Ecuyer generator is the one `sim()` sets in both of those R versions.

`exportData(program = "Mplus")` writes data files. The estimation scripts call `lavaan::sem`. The page quotes a lavaan warning and names `nlminb`. No Mplus output is on the OSF node. Had Mplus been the estimator, its default would fix the first loading at 1, and a default summary prints a convergence criterion of 0.00005 with at most 1,000 iterations. A negative residual variance is reported and the solution is still returned. The tables come from the lavaan scripts above. Mplus is not installed. `simulation/mgcfa_osf.inp` mirrors the multigroup string and was not executed.

The OSF node `q9yu8` was created on 2025-04-18. The files now on it were uploaded on 2025-09-20. They are the generation scripts, the lavaan analysis scripts, and the summary CSVs. There is no `sessionInfo`, no log, no `.Rhistory`, and no `.dat` replicate. The generation scripts call `set.seed(1313)` and pass seed 13131 to `exportData`. That call is commented out. One uncommented call, in the $N = 480$ condition-3 null script, names group sizes $(168, 168, 72, 72)$, the 35/15 split. The Monte Carlo here uses the printed shares.

GitHub, GitLab, and Zenodo searches, and a clone of the twenty public repositories under `jjeffries13`, are logged in `simulation/out/code_search.md`. Queries for the dissertation title, `osf/analysis/mgcfa_model.R`, and `set.seed(1313)` on that account returned no dissertation script. `jjeffries13/APA_2024` reuses the loading idiom and seed 13131 for a ten- and fifteen-indicator missing-data study. `Dynamic-Fit-Indices/DFI_Presentation.Rmd` calls `sessionInfo()` and sinks `DFIsessionInfo.txt`. That file is not in the repository. The commits run from March through October 2022, the file is dated 9 March 2022, and the working directory is under `/Users/jayjeffries/`. Two teaching repositories last changed in February and June 2023 carry a badge for R 4.2.1. A May 2022 file has a commented Windows path. The paths those years read sit on that Mac home directory. No 2023 or 2024 repository contains a saved `sessionInfo`, an `renv.lock`, a packrat file, or a package `DESCRIPTION`.

The dissertation text remains the source for R 4.2.2. A Mac path in other projects leaves the simulation machine unnamed, and it supplies no lavaan or simsem version. A BLAS difference would move a converged estimate in the last digits. It would not move a rejection rate by a percentage point. The 6.46 against 7.41, the average on the boundary, and the MIMIC counts short of 10,000 are what a second generator and the public record left in place. The author's `.dat` files, or a `sessionInfo` from the run that wrote the tables, are the check that would tell those three apart from this stream.

### 5.1 OpenMx on the unconverged MIMIC replicates

The question is whether the MIMIC replicates that nlminb left unconverged are bad datasets or an optimizer setting. The script is `simulation/openmx_check.R`. It replays the hand stream from `simulation/validate_10k.R` and the simsem stream from `simulation/env_miss.R`, keeps every replicate lavaan did not converge, and keeps 25 converged replicates in each of the eight cells as a control. OpenMx 2.22.11 [5] then fits the same free-intercept marker model. The CRAN build's default optimizer is SLSQP. NPSOL is not compiled into it. CSOLNP was run as a second optimizer.

The indicator covariance of a failed replicate has a condition number between about 3.6 and

5.4, and its smallest eigenvalue is between about 0.34 and 0.45. The nlminb attempt did not return a negative variance, and it did not raise the gradient warning. Those replicates do not look like a singular sample or a Heywood case in the data.

| Stream and cell | nlminb failed | SLSQP code 0 | CSOLNP code 0 | BFGS recovered |
|---|---|---|---|---|
| Hand, equal, true | 12 | 0 | 1 | 12 |
| Hand, equal, null | 35 | 1 | 1 | 35 |
| Hand, 45/5, true | 14 | 0 | 0 | 14 |
| Hand, 45/5, null | 48 | 0 | 0 | 48 |
| simsem, equal, true | 12 | 0 | 0 | 12 |
| simsem, equal, null | 36 | 0 | 1 | 36 |
| simsem, 45/5, true | 20 | 0 | 0 | 20 |
| simsem, 45/5, null | 49 | 0 | 1 | 48 |

OpenMx status code 0 is a converged fit. Code 1 did not occur. The other failures returned code 6, the RED status: the first-order conditions were not met. On the control replicates the largest absolute gap between a lavaan contrast and the OpenMx SLSQP contrast is $2.2 \times 10^{-5}$.

The same failed datasets were refit in lavaan with BFGS, with nlminb allowed 50,000 iterations, and with lavaan's simple starting values. The longer run and the simple starts converged none of them. BFGS converged 225 of the 226. Seven cells then have 10,000 converged replicates. The simsem null cell with shares 45/45/5/5 has 9,999. Counting those BFGS solutions, the contrast-2 rate in that cell is 6.51 in the simsem stream and 6.47 in the hand stream, against the printed 7.41. The printed convergence count of 10,000 is reached by changing lavaan's optimizer. The printed rejection rate is not. The dissertation names nlminb, and under nlminb the count stays short of 10,000.

Jeffries and Bovaird (2026), DOI `10.1177/01466216261452255`, was not obtained as a PDF. Unpaywall, Europe PMC, and the Wayback Machine had no PDF on 2026-10-27. The OSF node is the analysis record for that title.

## 6 Software

This section answers which versions produced the fits and the proofs. The session files are `simulation/out/session_info.txt` and `simulation/out/env_session.txt`.

The Monte Carlo ran in R 4.3.3 (2024-02-29) [1]. R Core Team (2024) is the citation for that installation. The project page is r-project.org.

Every fit used lavaan 0.6.17 [2]. Rosseel (2012) is *Journal of Statistical Software* 48(2), pages 1–36, 10.18637/jss.v048.i02. The package page is CRAN. The development repository is yrosseel/lavaan.

The ten-cell run used simsem 0.5.17. The redraw used simsem 0.5-16 [3]. The `citation` entry for that 0.5-16 install is Pornprasertmanit, Miller, Schoemann, and Jorgensen (2021), R package version 0.5-16. The package page is CRAN. The tarball installed for the redraw is `simsem_0.5-16.tar.gz`. The development repository is simsem/simsem.

The convergence check used OpenMx 2.22.11 [5]. Neale and coauthors (2016) is *Psychometrika* 81(2), pages 535–549, 10.1007/s11336-014-9435-8. The CRAN source is that version. On this R 4.3.3 the file `omxData.cpp` calls `Rf_isDataFrame`, which this R headers file does not declare, so that one call was compiled as `Rf_isFrame`. The default optimizer in the build is SLSQP.

The code identities were checked with Lean 4.22.0 [4]. The toolchain file `lean-toolchain` reads `leanprover/lean4:v4.22.0`. de Moura and Ullrich (2021) is *Automated Deduction – CADE 28*, Lecture Notes in Computer Science, Springer, pages 625–635, 10.1007/978-3-030-79876-5_37. The release is v4.22.0.

Mathlib [6] is the tag v4.22.0. `lake-manifest.json` pins commit `79e94a09`, and that commit is the tag. The mathlib Community (2020) is the *Proceedings of the 9th ACM SIGPLAN International Conference on Certified Programs and Proofs*, pages 367–381, 10.1145/3372885.3373824.

# 7 Assumptions

This section answers which choices were supplied here rather than read off the page.

## 7.1 Random-number stream

The 400-replicate screen used L'Ecuyer-CMRG, `set.seed(13131)`, and one drawn seed per cell. The 10,000-replicate run sets a separate fixed seed in each cell. The simsem redraw uses seed 13131 inside `sim()`. These are not Jeffries's replicates.

## 7.2 Package versions

The ten-cell Monte Carlo used R 4.3.3 and simsem 0.5.17. The miss-cell redraw installed simsem 0.5-16 and kept lavaan 0.6.17. R 4.2.2 was not installed. Versions are in `simulation/out/session_info.txt` and `simulation/out/env_session.txt`.

## 7.3 Convergence

A replicate counts as converged when `lavInspect(fit, "converged")` is true and the three contrast standard errors are finite. That flag is the one the OSF script reads from the summary header. In the simsem 0.5-16 redraw the two rules selected the same replicates.

## 7.4 The first intercept

The Monte Carlo generates from `i1 ~ 1*1`, the token in the OSF script and in the null model on the page.

## 7.5 The MIMIC intercept

The MIMIC Monte Carlo frees the latent intercept. The posted line, which fixes it at 0, does not recover −0.40.

# 8 Ambiguities that were resolved

This section answers which readings were settled by a file, a fit, and a command.

## 8.1 The minus signs on the generating contrasts

The question was whether the text extract had dropped the minus signs on −0.40, −0.20, and 0. Figure 2 is the rendered page. The crop script is `paper/figures/crop_pages.py`, command

`python3` paper/figures/crop_pages.py. The output crops are in paper/figures. The proof commit is `2dcc1c4`.

### 8.2 Which population produces −0.40

The question was whether the marker model on equal loadings 0.70 is the printed scale. The artifact is simulation/out/candidate_readings.csv, from `Rscript` simulation/candidate_readings.R `out`. The parsed string, with the first loading filled in as 1, matches. The equal-loading marker reading does not.

### 8.3 The public analysis script

The question was which identification produced Tables 4.1 and 4.3. The artifact is osf/analysis/mgcfa_model.R, retrieved 2026-10-27. The multigroup summary matches Table 4.1 to the printed digits. The Monte Carlo of that string is simulation/out/validate_10k.csv, from `Rscript` simulation/validate_10k.R `10000 out`, scored by `Rscript` simulation/score_gaps.R simulation/out/validate_10k.csv simulation/out/gap_scores_10k.csv.

## 9 Code and source availability

This section answers where each source was looked up, and what came back.

| Place | Date | Result |
|---|---|---|
| Dissertation PDF, Appendix A | 2026-10-27 | Generation syntax. sha256 `a21df98c...fd469b3` |
| OSF node `q9yu8` | 2026-10-27 | Scripts and summaries. No `.dat` files, no `sessionInfo` |
| DOI `10.1177/01466216261452255` | 2026-10-27 | No PDF from the publisher, Unpaywall, Europe PMC, or Wayback |
| lavaan 0.6.17 | 2026-10-27 | Used for every fit |
| simsem 0.5-16, CRAN archive | 2026-10-28 | Used in simulation/env_miss.R. Generator is `lavaan::simulateData` |
| GitHub, GitLab, Zenodo | 2026-10-28 | No dissertation script. Log in simulation/out/code_search.md |
| OpenMx, MplusAutomation, Mplus | 2026-10-27 | Not installed |

The dissertation file was retrieved from the Wayback `id_` object dated 2025-05-11, snapshot `20250511025154`. The help request filed before the OSF node was found is https://github.com/alectionary/vibe-math/issues/33. The fetch request for the 2026 article is https://github.com/alectionary/vibe-math/issues/34.

## 10 Next steps

The check that would close the three open comparisons is Jeffries's `.dat` files, or a `sessionInfo` from the run that wrote Tables 4.1 through 4.5. An R 4.2.2 binary would test whether the L'Ecuyer draws differ from R 4.3.3. The 2026 PDF would show whether the journal version uses these tables.

## 11 Artifacts

- Code identities. File LatentContrasts/Orthogonality.lean. Command: `lake build`. Commit 2dcc1c4.
- Moment fits. File simulation/out/candidate_readings.csv. Command: `Rscript simulation/candidate_readings.R out`.
- Large-sample MIMIC. File simulation/out/mimic_large.csv. Command: `Rscript simulation/mimic_large.R out`. Seed 7.
- Four-hundred-replicate screen. File simulation/out/validate_gaps.csv. Command: `Rscript simulation/validate_gaps.R 400 out`. Scores: `Rscript simulation/score_gaps.R simulation/out/validate_gaps.csv simulation/out/gap_scores.csv`.
- Ten-thousand-replicate run. File simulation/out/validate_10k.csv. Seeds: simulation/out/seeds_10k.csv. Command: `Rscript simulation/validate_10k.R 10000 out`. Scores: `Rscript simulation/score_gaps.R simulation/out/validate_10k.csv simulation/out/gap_scores_10k.csv`.
- OpenMx convergence check. File simulation/out/openmx_summary.csv. Command: `Rscript simulation/openmx_check.R out`.
- simsem 0.5-16 redraw. File simulation/out/env_miss.csv. Command: `Rscript simulation/env_miss.R 10000 out`.
- Code search. File simulation/out/code_search.md.
- Page crops. Directory paper/figures. Command: `python3 paper/figures/crop_pages.py`.

The dissertation PDF is local and gitignored. Its sha256 is `a21df98c20401491e6b4a1d00a21b6f7cee5551a00c04a89fb414ec6cfd469b3`.

## References


[1] R Core Team. *R: A Language and Environment for Statistical Computing*. R Foundation for Statistical Computing, Vienna, Austria, 2024. https://www.R-project.org/.

[2] Yves Rosseel. lavaan: An R package for structural equation modeling. *Journal of Statistical Software* 48(2) (2012), 1–36. 10.18637/jss.v048.i02.

[3] Sunthud Pornprasertmanit, Patrick Miller, Alexander Schoemann, and Terrence D. Jorgensen. *simsem: SIMulated Structural Equation Modeling*. R package version 0.5-16, 2021. https://CRAN.R-project.org/package=simsem.

[4] Leonardo de Moura and Sebastian Ullrich. The Lean 4 theorem prover and programming language. In *Automated Deduction – CADE 28*, Lecture Notes in Computer Science, Springer, 2021, pages 625–635. 10.1007/978-3-030-79876-5_37.

[5] Michael C. Neale, Michael D. Hunter, Joshua N. Pritikin, Mahsa Zahery, Timothy R. Brick, Robert M. Kirkpatrick, Ryne Estabrook, Timothy C. Bates, Hermine H. Maes, and Steven M. Boker. OpenMx 2.0: Extended structural equation and statistical modeling. *Psychometrika* 81(2) (2016), 535–549. 10.1007/s11336-014-9435-8.

[6] The mathlib Community. The lean mathematical library. In *Proceedings of the 9th ACM SIGPLAN International Conference on Certified Programs and Proofs*, 2020, pages 367–381. 10.1145/3372885.3373824.

**Table 4.5**
*Initial Model Convergence Rates by Condition & Model*

| *N* | Condition | MIMIC Frequency Converged (%) | MGCFA Frequency Converged (%) |
|---|---|---|---|
| Significance Models | | | |
| 240 | 1 | 10000 (100%) | 9886 (98.86%) |
| | 2 | 10000 (100%) | 9811 (98.11%) |
| | 3 | 10000 (100%) | 8929 (89.29%) |
| 480 | 1 | 10000 (100%) | 9989 (99.89%) |
| | 2 | 10000 (100%) | 9994 (99.4%) |
| | 3 | 10000 (100%) | 9805 (98.05%) |
| 960 | 1 | 10000 (100%) | 10000 (100%) |
| | 2 | 10000 (100%) | 10000 (100%) |
| | 3 | 10000 (100%) | 9986 (99.86%) |
| Null Models | | | |
| 240 | 1 | 10000 (100%) | 9932 (99.32%) |
| | 2 | 10000 (100%) | 9911 (99.11%) |
| | 3 | 10000 (100%) | 9675 (95.75%) |
| 480 | 1 | 10000 (100%) | 9998 (99.98%) |
| | 2 | 10000 (100%) | 10000 (100%) |
| | 3 | 10000 (100%) | 9999 (99.99%) |
| 960 | 1 | 10000 (100%) | 10000 (100%) |
| | 2 | 10000 (100%) | 10000 (100%) |
| | 3 | 10000 (100%) | 10000 (100%) |

Figure 4: Table 4.5. At $N = 240$ and condition 3 the multigroup significance model converged in 8,929 of 10,000 replicates. The null block of the same table prints 9,675 and 95.75%.